

Index Number:

Name : _____

CT Group : 07S _____

HWA CHONG INSTITUTION**JC2 PRELIMINARY EXAMINATIONS
H1 BIOLOGY 8875
PAPER 2****11 Sept 2008****2 hrs*****INSTRUCTIONS TO CANDIDATES***

There are **2** question booklets (**I** to **II**) to this paper. Write your **name**, **CT** and **Index Number** on the cover pages of the booklets in the spaces provided. Write in blue or black pen.

SECTION A

This section contains **4** questions. Answer **ALL** questions.

Write your answers in the spaces provided on the question paper.

SECTION B

Answer **ONE** question.

Your answer to **Section B** must be in continuous prose, where appropriate.

Write your answers on the separate answer paper provided.

At the end of the examination,

1. write the number of the **Section B** question you have answered in the opposite grid. **A NIL return is required.**
2. submit all question booklets and answer sheets to **Section B** separately.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
Total	/ 60

SECTION A: STRUCTURED QUESTIONS

QUESTION 1

Fig. 1.1 and **Fig. 1.2** show the structures of glycogen and triglyceride respectively. Both glycogen and triglyceride are metabolic fuels.

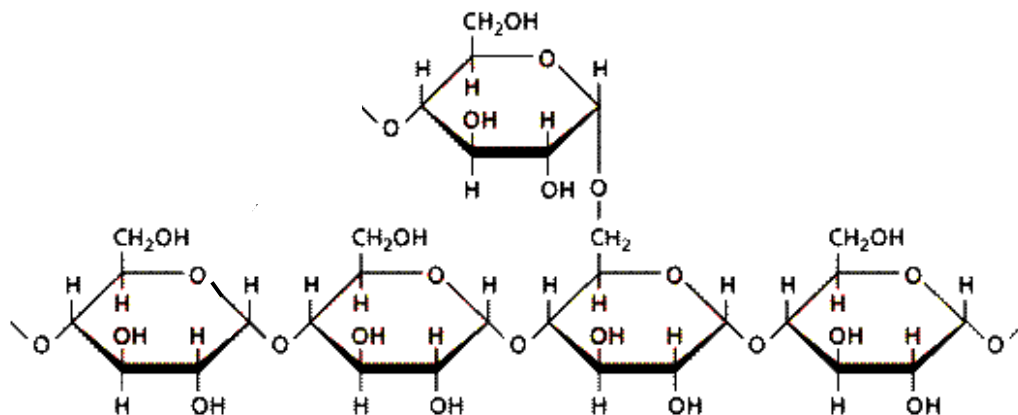


Fig. 1.1

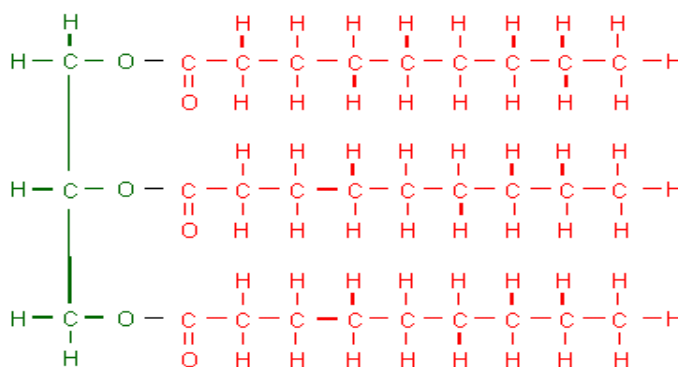


Fig. 1.2

With reference to **Fig. 1.1** and **Fig. 1.2**,

- (a) state one similarity and two differences in the structure of glycogen and triglyceride. [3]

Similarity: _____

Differences

1. _____



2. _____

(b) which metabolic fuel is a better form of energy storage in the human body? Give a reason for your answer. [2]

Fig. 1.3 shows some steps involved in glycolysis and the Krebs cycle. Some ATP is made directly. Hydrogen is also released and this can result in the production of more ATP.

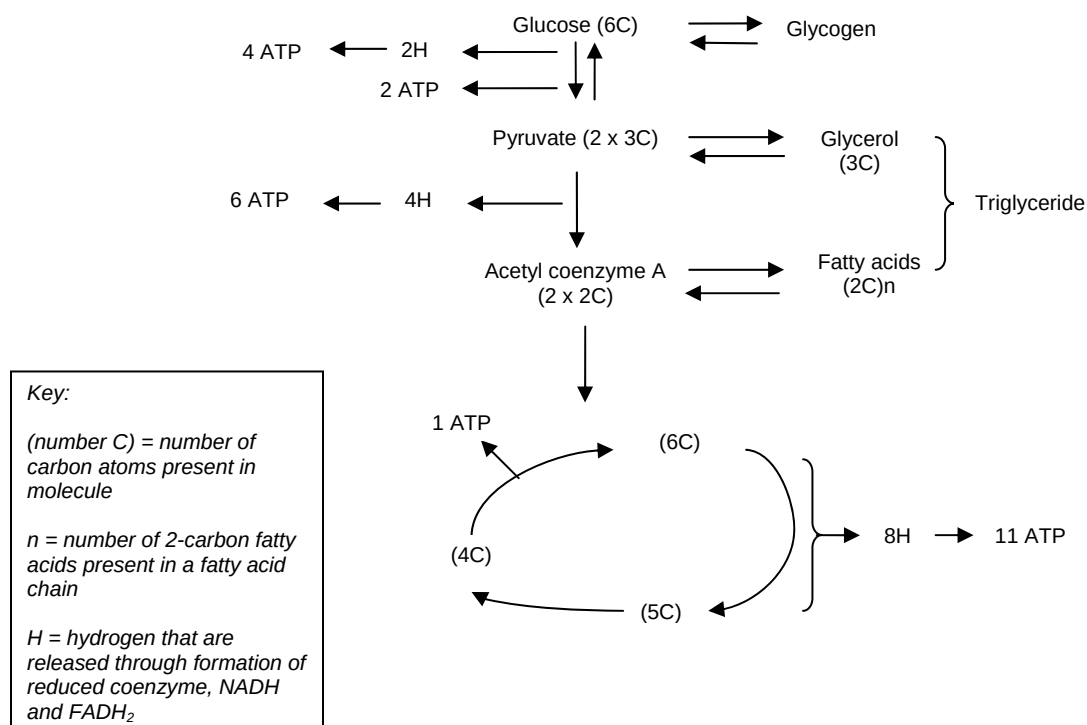


Fig. 1.3

- (b)** Describe how the hydrogen released during glycolysis and the Krebs cycle results in the production of ATP. [2]

Similar to glycogen, triglycerides can be respired. The first step is to hydrolyse each triglyceride molecule. Each fatty acid formed is broken down into acetyl CoA molecules. The acetyl CoA molecules then enter the Krebs cycle.

- (c)(i)** Using the information from **Fig. 1.3**, calculate the number of ATP molecules produced from the complete oxidation of a 16-carbon fatty acid. Show your answer in the space provided below. [2]

- (ii)** Suggest why fatty acids can only be respired under aerobic conditions. [2]

- (iii)** Glucose catabolism is spread out over many reactions as shown in **Fig. 1.3**. Suggest an advantage to this. [1]

[Total: 12 marks]

QUESTION 2

The organisation of bacterial genome differs from that of eukaryotic cells in several ways. **Fig. 2.1** below shows an electron micrograph of a dividing pair of bacterial cells.

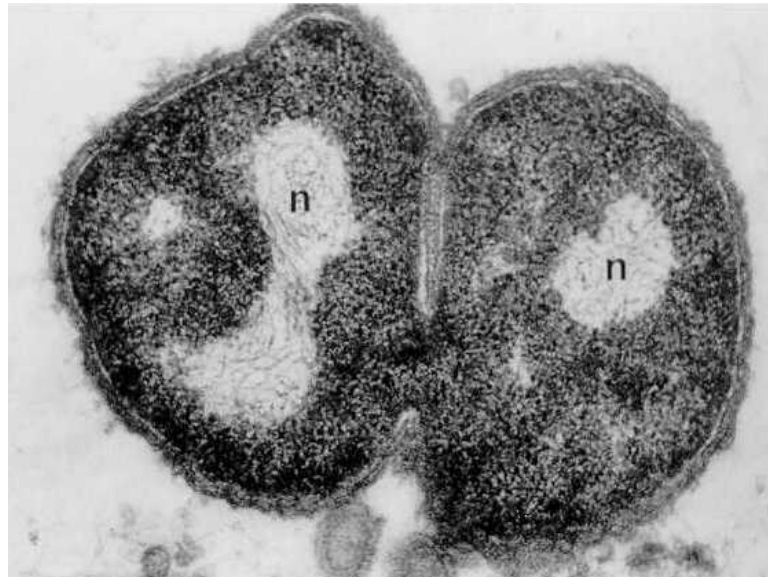


Fig. 2.1

- (a)(i)** With reference to **Fig. 2.1**, state one feature that is present in the cytosol of the bacterial cells that is not found in eukaryotic cells. [1]

- (ii)** Name one organelle that is present in both bacterial and eukaryotic cells and state its function. [2]

A group of researchers has discovered a new bacteria strain, and has successfully isolated its DNA using Pulsed Field Gel Electrophoresis (PFGE), an approach that uses alternating electric fields to separate DNA molecules on an agarose gel. The DNA makeup of the the bacteria is shown in Table 2.1

DNA Type	Molecular Weight	Occurrence
I	6.7 Mb	1 copy
II	300 kb	several copies

Table 2.1

With reference to **Table 2.1**,

(b)(i) identify **DNA Type I** and **II**. [1]

DNA Type I : _____

DNA Type II : _____

(ii) explain your answers to **(b)(i)** and state one function of **DNA Type II**. [3]

(c) Discuss the differences in the genes present in **DNA Type I** and **Type II**. [2]

Many genes in bacteria are expressed constitutively, that is, they are always turned "on". However, some bacterial genes are organised adjacent to each other, and they are regulated coordinately, an example of which is the *lac* operon in *Escherichia coli*. **Fig. 2.2** shows the structure of a *lac* operon .

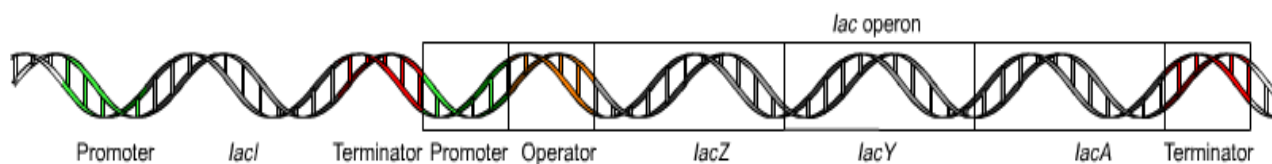


Fig. 2.2

- (d) With reference to the **Fig. 2.2**, explain the effect of a single base substitution at the operator region on the *lac* operon. [2]

In an experiment, a batch of *Escherichia coli* is divided into two tubes with growth medium containing glucose, but lactose is added to one of the tubes.

- (e)(i) State any difference there might be in the growth rate between the two tubes. [1]

- (ii) Explain your answer to (e)(i). [2/3]

[Total: 15 marks]

BOOKLET II

SECTION A: STRUCTURED QUESTIONS (Continued)

QUESTION 3

There are several subspecies of the mouse, *Mus musculus*, living throughout the world. A study was conducted to study their spread from an ancestral population. **Fig. 7.2** shows the results of sequencing mitochondrial DNA from 10 mice from geographically diverse locations. Dashes indicate bases that are identical to the top sequence.

England	AAAGCAGAAGAAATAGATGCAAAGAAGAAGAGTTCAA
Japan	--GCAT-----C-----AATT
South Africa	-----C-----TA-----
India	GG-CG--C-----GGG-GG-----CGGG
China	TTT--T-----TT--T-----GGTT
United States	-----C-----T-----
Philippines	TTT--T-----TC--T-----GGTT
New Zealand	-----C-----C-TA-----
Russia	--GCAT-----CC-----A-TT
Yemen	-----C-----

Fig. 7.2

(a)(i) Draw a box in **Fig. 7.2** to include the entire region that most likely codes for an important structure in the mitochondrion. [1]

(ii) Explain your answer to (a)(i). [2]

(iii) With respect to **Fig. 7.2**, identify the subspecies which is most closely related to the mouse found in England. Give a reason for your answer. [2]



(iv) Explain how the large number of mouse subspecies might have evolved. [2]

[Total: 7 marks]

QUESTION 4

[Total: 11 marks]

--- End of Section A ---



SECTION B: FREE RESPONSE QUESTIONS

Answer **ONE** of the following two free-response questions. A **nil return** is required.

Your answers must be in continuous prose, where appropriate.

Answers should be illustrated by large, clearly visible diagrams, where appropriate.

QUESTION 5

- (a) Describe the structural features of a named membranous organelle that is involved in ATP synthesis in photosynthesis. [6]
- (b) Compare the two main pathways in which an excited electron can undergo during the light-dependent stage of photosynthesis. [6]
- (c) With reference to the term 'limiting factor', discuss how two named factors have effects on photosynthesis. [8]

QUESTION 6

- (a) Describe the structural features of two named nucleic acids involved in protein synthesis. [8]
- (b) Compare the two processes that have to take place for protein synthesis. [8]
- (c) Discuss two named factors that would affect the rate of protein synthesis. [4]

--- End of Section B ---

